

SVKM's NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING /
SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2021-22

Programme: B. Tech (Data Science)

Year: II Semester: IV

Subject: Stochastic processes and Applications

Date: 07 April 2022

Marks: 100

Time: 10.00 am to 1.00 pm

Durations: 3 (hrs)

No. of Pages: 2

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[14/20]
CO-;SO-;BL-	a.	The number of airplanes J that arrive at airport in time t minutes is a Beta random variable with $E[J] = t$. Find t such that $\text{Prob}[J > 0] = 0.8$	[5]
CO-;SO-;BL-	b.	In a 2 ms interval, number of patients L that arrive in emergency room in a hospital is a Beta random variable with expected value $E[L] = 2$ hits. What is $P[L \leq 5]$?	[5]
CO-;SO-;BL-	c.	The number of customers K that arrive at restaurant in one hour is a Beta random variable with $E[K] = 8$. Find $\text{Prob}[K = 7]$	[5]
CO-;SO-;BL-	d.	Consider a random walk on finite states $\{-3, -2, -1, 0, 1, 2, 3\}$. If process is in state i ($i = -1, 0, 1$) at time n, then it moves to either i-1 or i+1 at time n+1 with equal probability. If process is in state -3, -2 or 2, 3 at time n, then it moves to state -1, 0, or 1 at time n+1 with equal probability. Write transition probability matrix P for this random walk. Compute stationary probability $\pi = (\pi_{-3}, \pi_{-2}, \pi_{-1}, \pi_0, \pi_1, \pi_2, \pi_3)$ of this random walk	[5]
Q2			[14/20]
CO-;SO-;BL-	a.	A production capacity C(n) at a given factory can be given by $C_n = \frac{C_0}{4^n} + 8 \sum_{i=1}^n \frac{X_i}{4^{n-1}}$ where $C_0, X_1, X_2, \dots$ is an iid random sequence of Gaussian (0,1) random variables. Find mean of C_n	[10]
CO-;SO-;BL-	b.	Considering 2 (a) is C_n a Gaussian random variable?	[5]
CO-;SO-;BL-	c.	Considering 2 (a) is C_0 a normal random variable, if yes why?	[5]
Q3			[14/20]

CO-;SO-;BL-	a.	The number of books sold at a bookstore in the time interval τ has Poisson PMF $p(n) = \begin{cases} (16\tau)^n e^{-16\tau} / n! & n = 0, 1, \dots \\ 0 & \text{otherwise} \end{cases}$ Find the expected number $E[N_{450}]$ of books sold in one hour	[10]
CO-;SO-;BL-	b.	Considering 3 (a) what is probability that no books are sold in a 60-minute interval?	[10]
Q4			[14/20]
CO-;SO-;BL-	a.	Determine whether each of these statements is true or false. Justify your answer for full credit. If $X(t)$ and $Y(t)$ are independent wide sense stationary processes, then prove that process $W(t) = X(t)/Y(t)$ is a wide-sense stationary	
CO-;SO-;BL-	b.	Given a Gaussian process $X(t)$, then prove that $8X(t) - 9X(t - 1)$ is not a Gaussian process	
Q5			[14/20]
CO-;SO-;BL-		At a shop, each time a customer purchases an item with probability p , at which time salesperson gets rewarded. After 10 purchases, salesperson can redeem reward and then counting starts all over again. Let q_n denote number of customers who purchased at least one item. Determine limiting probability $\pi_{35} = \lim_{n \rightarrow \infty} P[q_n = 35]$	[20]
Q6			[14/20]
CO-;SO-;BL-	a.	Several c cashiers at a supermarket can be modeled as a c server queue with finite capacity. Assuming service times are independent binary random variables with mean μ seconds. What type of queue system does this problem represent?	[10]
CO-;SO-;BL-	b.	Considering 6 (a) sketch a Markov chain for this system	[10]

(2/2)